

Changrui (Ryan) Li

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EDUCATION

Carnegie Mellon University

Pittsburgh, PA

B.S. Computer Science (Machine Learning & Security and Privacy concentrations), 3.89 QPA

Present

HONORS & AWARDS

Dean's List High Honors, Carnegie Mellon University

Fall 2025, Spring 2026

AIME Qualifier (4x) | USACO Silver Division

2021 – 2024

NSDA Debate: District Champion, State 3rd Place

2025

EPS Innovating Wisely Award (4 of 80 graduates)

Jun 2025

TECHNICAL SKILLS

Languages: Java, Kotlin, Nix, Python, Go, Rust, C#, TypeScript, HTML/CSS, SQL(PostgreSQL), C, R

Frameworks: PyTorch, LangGraph, PydanticAI, Flask, Django, Node.js, React

Developer Tools: Git, Linux/NixOS, AWS (AgentCore), PyCharm, VSCode, IntelliJ, GCP, Microsoft Azure

EXPERIENCE

AI Developer and Data Scientist, MaxMyPoint and MaxFHR | Python, MCP Servers Dec 2024 – Present

MaxMyPoint: a website that helps travelers find and maximize hotel award redemptions.

- Developing a LangGraph + PydanticAI + AWS AgentCore chatbot to assist over 56,000 users in effectively navigating the website and suggesting optimal travel plans.
- Prototyped an MCP server to enable the chatbot to perform direct operations for users, gated by account level and question relevance.
- Processing PostgreSQL data to track hotel alert success rates by type, hotel, and time, using AI models to validate code correctness and suggest improvements.
- Integrated persistent memory tracking into the chatbot to personalize travel recommendations based on user history and past question patterns.

Research Assistant, CyLab | Kotlin, Android SDK, PostgreSQL

Oct 2025 – Present

- Traced a silent messaging regression to an undocumented upstream change and rebuilt all affected modifications to preserve study validity.
- Implemented a secure module within an open-source SMS application to facilitate real-time scam detection.
- Engineered an encrypted data pipeline to securely transmit user interaction data for real-time AI analysis.
- Led all technical development for the platform, translating research hypotheses and data requests into robust Android features and managing the integration of AI-generated responses into the mobile experience.

DevOps and Internal Tooling Tech Lead, ScottyLabs | Nix, Go, Rust, TypeScript

Sep 2025 – Present

- Deployed and maintained a Matrix homeserver with custom bridge integrations for bidirectional real-time communication mirroring across Slack, Discord, and Matrix for the entire organization.
- Standardized deployment tooling across 8+ projects, providing reproducible builds, branch-based preview environments, and unified secret management.
- Responded immediately to VM crashes causing outages for 3,000+ users, restoring service and diagnosing root causes to prevent recurrence.
- Contributed to the governance automation system that enforces team membership, repository permissions, and role sync from a single source of truth across 6 platforms.

PROJECTS

Bluetooth R&D Intern, Google | Java, C#, .NET, Android, Unity, Bluetooth | [GitHub](#)

Jun 2024 – Sep 2024

- Extended Google's open-source XDTK toolkit with a Bluetooth transport layer enabling Android devices to stream real-time sensor data to Unity applications without Wi-Fi, after evaluating and ruling out a BLE-based approach.
- Designed a custom packet protocol with flow control and boundary reassembly to achieve reliable high-frequency sensor streaming over Bluetooth.
- Built a native Windows Bluetooth DLL enabling Unity to discover and manage multiple simultaneous device connections with stable reconnection handling per device.